

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.397976696130359, median 0.3343657379478156, std: 0.3150660934224557
Gyroscope error (imu0): mean 4.52416713011731, median 2.437447646184137, std: 9.385727742673138
Accelerometer error (imu0): mean 8.580545605224904, median 3.827641379954786, std: 18.12892216338023

Residuals

Reprojection error (cam0) [px]: mean 0.397976696130359, median 0.3343657379478156, std: 0.3150660934224557
Gyroscope error (imu0) [rad/s]: mean 0.002595254472550537, median 0.0013982235234539478, std: 0.005384052180601584
Accelerometer error (imu0) [m/s²]: mean 0.060794366421243015, median 0.027119374837934248, std: 0.128445950587526

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[-0.03108649 -0.99887021 -0.03594357 0.03716061]
 [ 0.00993475 0.03565039 -0.99931494 0.08207563]
 [ 0.99946732 -0.03142228 0.00881528 -0.11777488]
 [ 0.         0.         0.         1.         ]]
```

T_ic: (cam0 to imu0):

```
[[-0.03108649 0.00993475 0.99946732 0.11805193]
 [-0.99887021 0.03565039 -0.03142228 0.03049185]
 [-0.03594357 -0.99931494 0.00881528 0.0843933 ]
 [ 0.         0.         0.         1.         ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
-0.0769307960225713

Gravity vector in target coords: [m/s²]

[0.17071027 1.48068519 -9.69261844]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [158.36275150387033, 158.59395308258087]

Principal point: [159.11277719780813, 159.18272823405007]

Distortion model: radtan

Distortion coefficients: [-0.0028376906083280792, -0.001327176938226464, -0.0004723408166781164, -0.0018829704422051435]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

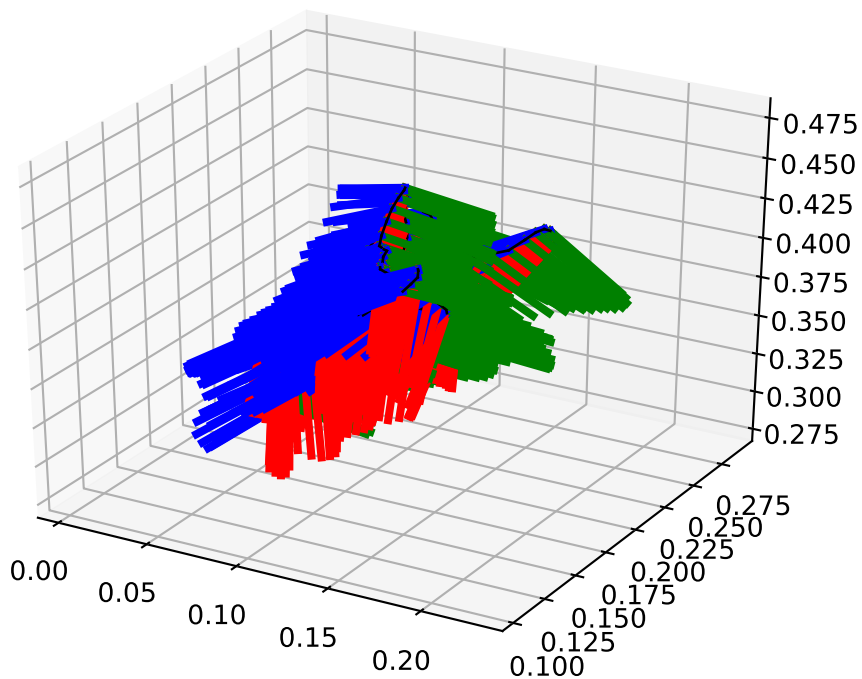
[0. 1. 0. 0.]

[0. 0. 1. 0.]

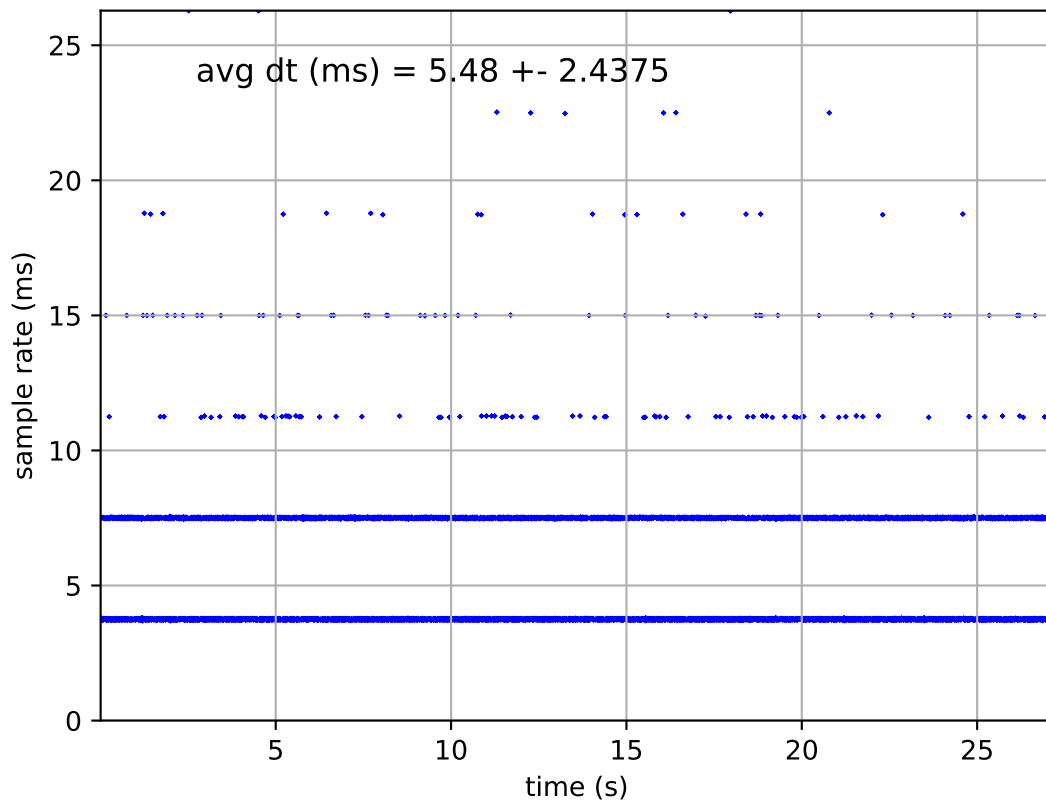
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

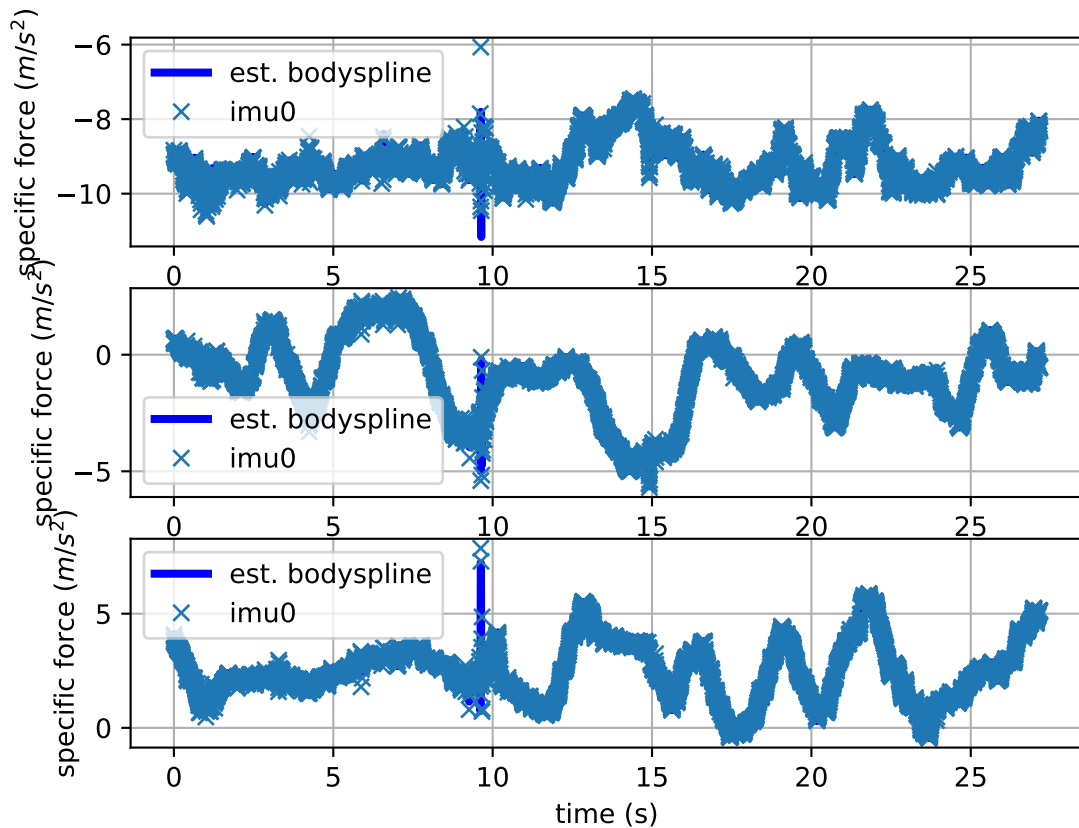
imu0: estimated poses



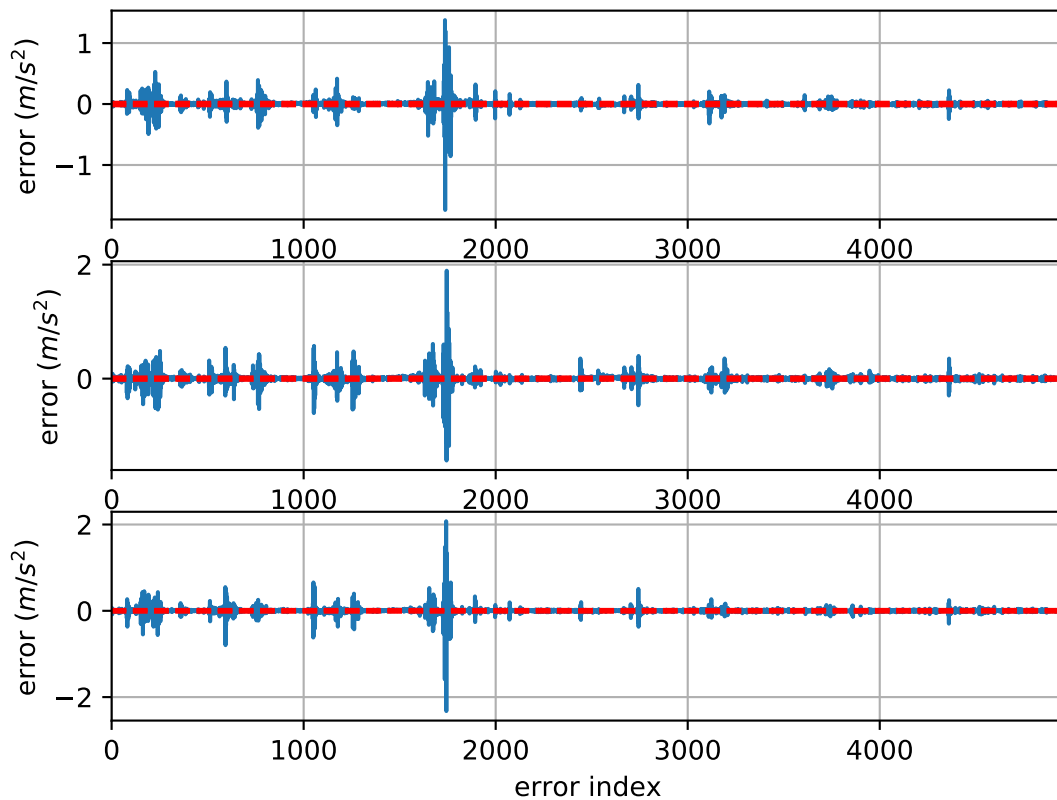
imu0: sample inertial rate



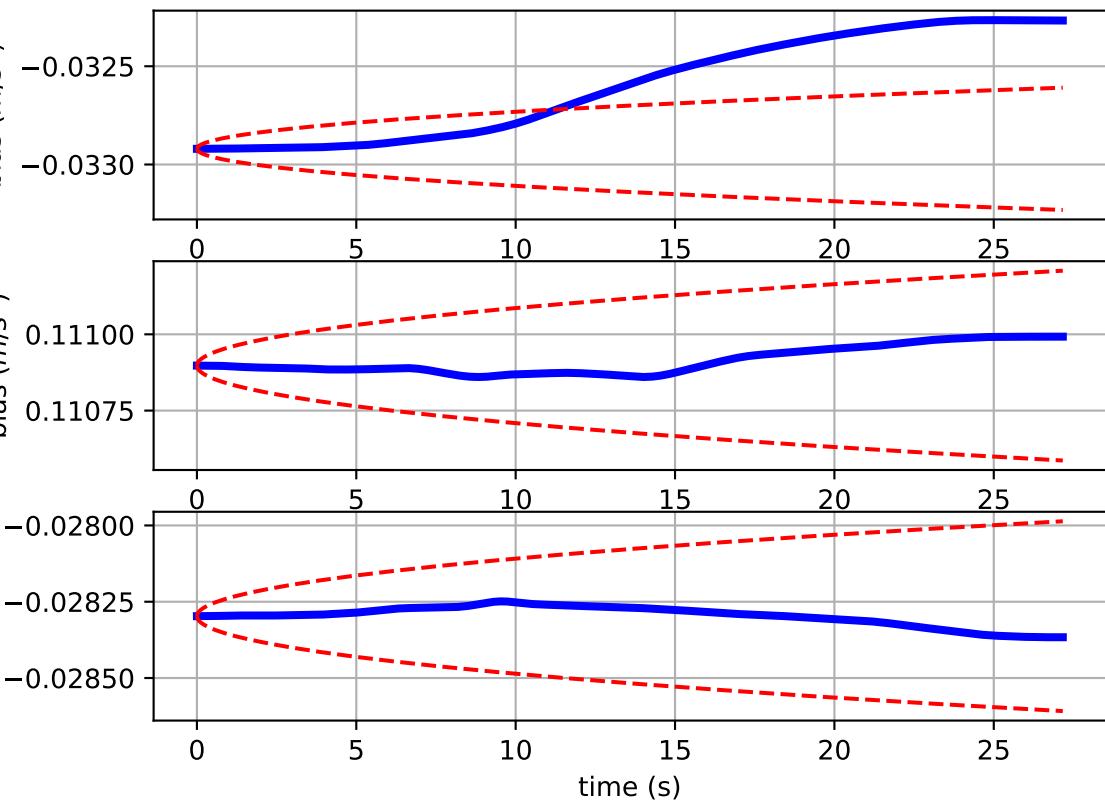
Comparison of predicted and measured specific force (imu0 frame)



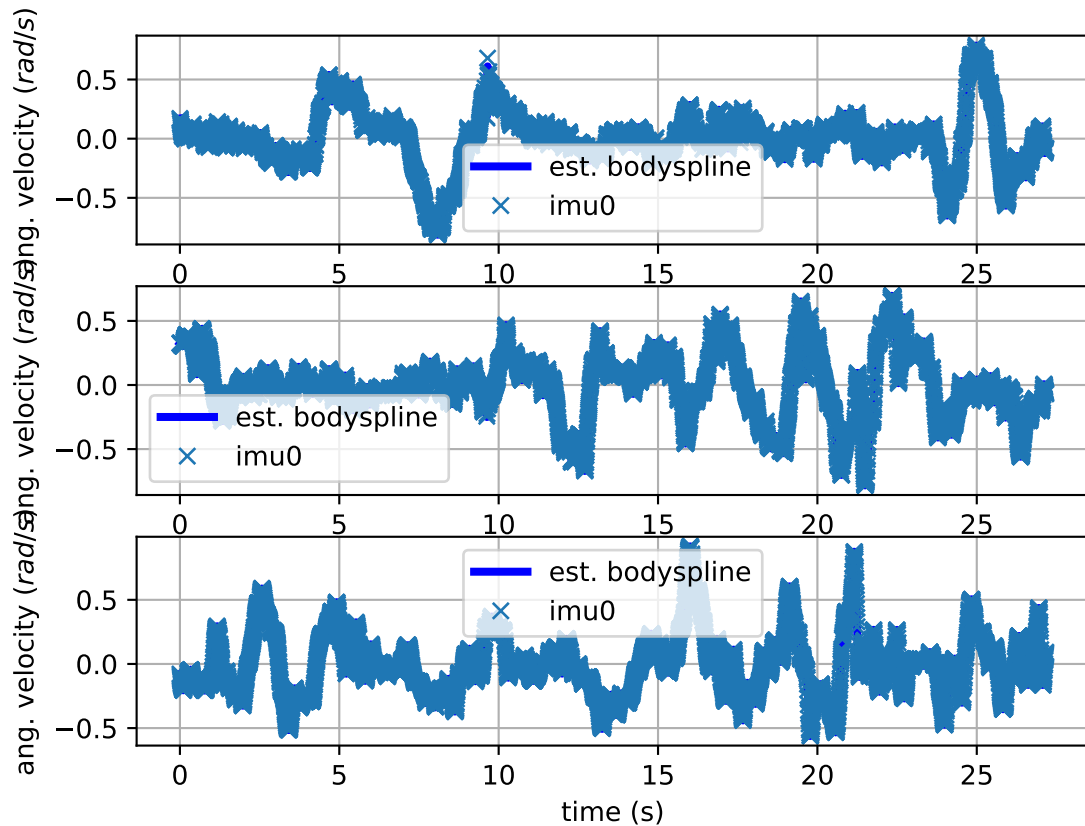
imu0: acceleration error



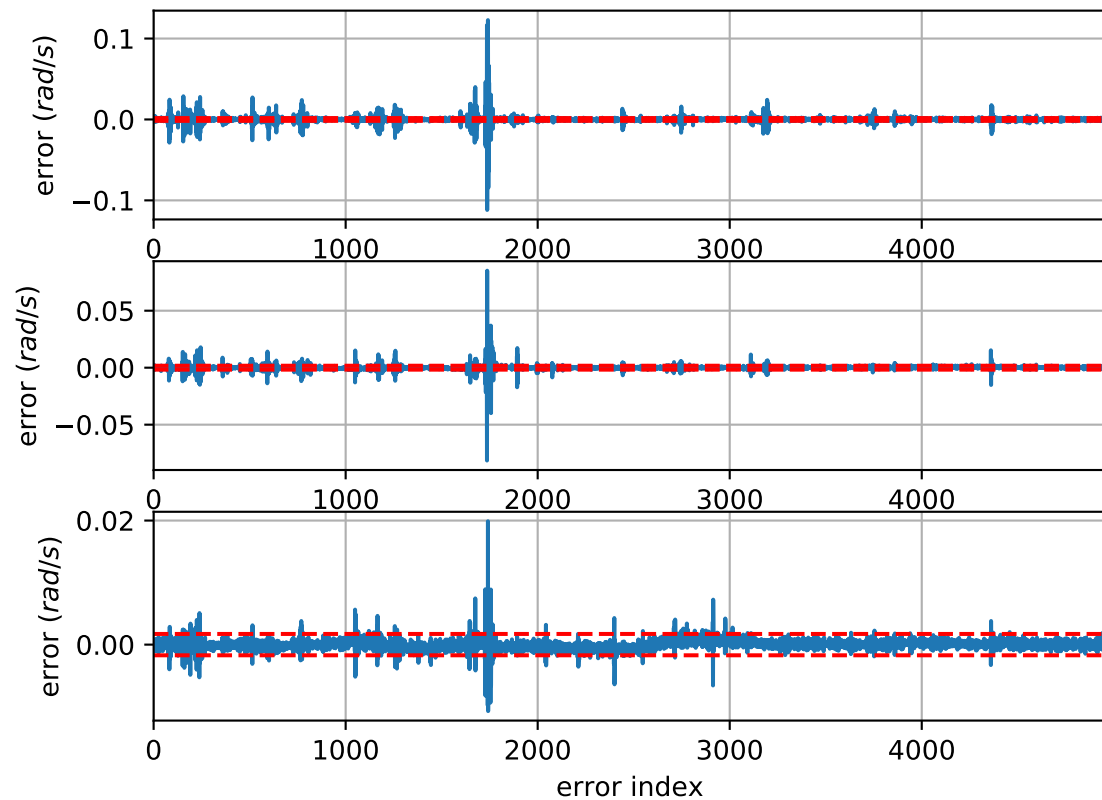
imu0: estimated accelerometer bias (imu frame)



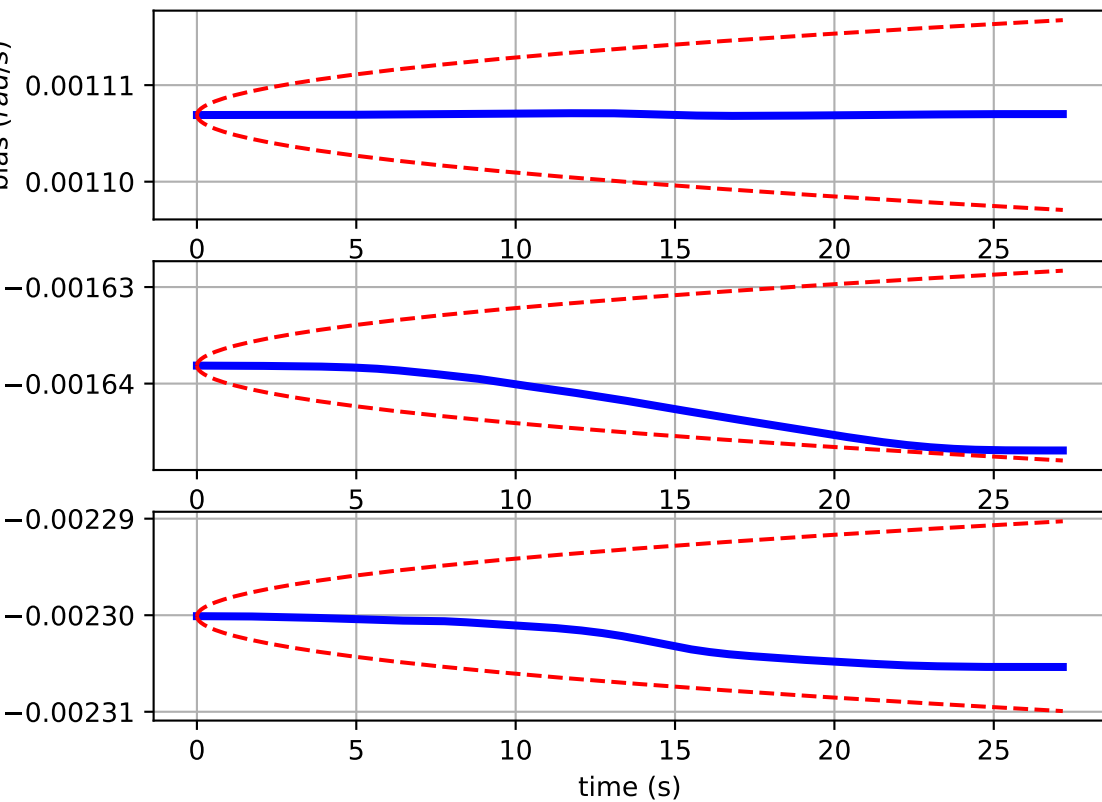
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

